CURRENT ISSUES OF SPACE LAW BEFORE THE UNITED NATIONS +

*Stephen Gorove**

The United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) has been the main instrument of the United Nations responsible for the preparation and drafting of major international resolutions and agreements pertaining to the law of outer space.[1] While the Committee, with the assistance of its two Subcommittees (the Scientific and Technical Subcommittee and the Legal Subcommittee), has had a remarkable track record based on consensus during its short-lived existence of less than a quarter of a century, it was unable to arrive at a general agreement with respect to principles governing direct television broadcasting by satellites which were adopted by the U.N. General Assembly by majority vote last fall.[2] Additionally, the Committee has not been able to reach consensus on three items still on its agenda:

A. Legal implications of remote sensing of the earth from space, with the aim of formulating draft principles;

B. Consideration of the possibility of supplementing the norms of international law relevant to the use of nuclear power sources in outer space and;

C. Matters relating to the definition and/or delimitation of outer space and outer space activities, bearing in mind, *inter alia,* questions relating to the geostationary orbit.

The purpose of this presentation is to identify the basic legal issues in the three agenda items and explain the rationale underlying the different approaches and suggested solutions.

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*International Astronautical Federation representative before the U.N. Committee on the Peaceful Uses of Outer Space (UNCOPUOS) and UNISPACE 82; Corresponding member, International Academy of Astronautics; Chairman, Editorial Board and Advisors, Journal of Space Law and Professor of Law, University of Mississippi Law center.

+ The views expressed in this article are those of the author and do not necessarily represent those of any organization with which he is connected.

[1]There has been a growing literature on the work and accomplishments of the U.N. Committee on the Peaceful Uses of Outer Space. *See,* for instance, Galloway, *Consensus Decisionmaking by the United Nations Committee on the Peaceful Uses of Outer Space,* 7 J. SPACE L. 3 (1977): Gorove, *The 1980 Session of the U.N. Committee on the Peaceful Uses of Outer Space: Highlights of Positions on Outstanding Legal Issues,* 8 J. SPACE L. 174 (1980). Hosenball, *The United Nations Committee on the Peaceful Uses of Outer Space: Past Accomplishments and Future Challenges,* 7 J. SPACE L. 95 (1977): Jankowitsch, *Contributions of the United Nations Committee on the Peaceful Uses of Outer Space: An Overview,* 5 J. SPACE L. 7 (1977): Jasentuliyana, *Report on the Work of United Nations Committee on the Peaceful Uses of Outer Space in 1981,* 9 J. SPACE L. 171 (1981): *idem, The Work of the United Nations Committee on the Peaceful Uses of Outer Space in 1982,* 10 J. SPACE L. 41 (1982).

[2]U.N. Gen. Assembly Res. 37/92 (Dec. 10,1982). Text reproduced in 10 J. SPACE L. 252 (1982).

A. Legal Implications of Remote Sensing

Consideration of the legal implications of remote sensing of the earth from space has been on the agenda of both the Scientific and Technical Subcommittee and the Legal Subcommittee for a number of years. A set of draft principles has been developed, many portions of which are still in square brackets indicating that no consensus on them has been achieved.[3] One of the major issues with respect to remote sensing has been the determination of conditions under which decisions on the acquisition and use of satellite data could be made. More specifically, the basic question continued to be whether there should be unlimited freedom to disseminate data and information gained by remote sensing or whether it was necessary to require consent by the sensed state and impose restrictions.[4]

In course of the discussions many less developed countries opposed the uncontrolled dissemination of detailed space imagery of a foreign state since in their view such dissemination damaged the sensed state's interests and caused international friction.[5] Therefore, in their view it was essential to protect the legitimate rights and interests of the sensed state against the misuse of data and information about its national territory. Otherwise the principle of state sovereignty, which was the keystone of international law, would be violated.[6] In this connection, it was noted that

> due respect for the sovereignty of sensed state, implied the observance of two fundamental principles: first, that the sensed state should have timely and preferential access at nominal cost (reasonable terms) to data obtained by remote sensing of its territory and, second, that such data should not be disseminated to third states without its prior authorization.[7]

With respect to the first principle, the view was expressed that it was unproductive to establish a dividing line between primary data and analyzed information as a basis for the granting of access to such data.[8] However, it was also stated that due regard had to be given to the efforts involved in producing such analyses[9] and that proprietary rights had to be taken into account.[10]

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[3]For a text of the Draft Principles incorporating the changes made at the Twenty-Second Session of the Legal Subcommittee, *see* U.N. Doc. A/AC. 105/320, pp. 16-21 (13 April 1983), reproduced *infra*, pp. 165-169.

[4]Docs. A/AC.105/C.2/SR.382, p. 4 (1983); A/AC.105/C.2/SR.383, p. 8 (1983).

[5]Doc. A/AC.105/C.2/SR.383, p. 2.

[6]*Id.* at 8.

[7]Doc. A/AC.105/C.2/SR.387, p. 4 (1983).

[8]Doc. A/AC.105/C.2/SR.383, p. 7 (1983).

[9]Doc. A/AC.105/C.2/SR.387, p. 4 (1983).

[10]Doc. A/AC.105/C.2/SR.384, p. 6 (1983).

Concerns about free dissemination of data arose from perceptions of new remote sensing techniques which have made it possible to evaluate both agricultural and mineral resources. In this respect it was noted that

> a country with the right technology could acquire a precise picture of the current state and economic potential of another country by remote sensing. The data which could be obtained included natural resources, industrial installations, means of communication and the like and that was the reason why legal principles had to be formulated to regulate properly the problems currently posed by remote sensing.[11]

The theory was also advanced that unlimited freedom to disseminate data and information could place the space powers in an even more advantageous position in international markets by cornering data on resources of other countries[12].

While some less developed countries were of the view that the concept of permanent sovereignty over natural wealth and resources applied to all data and information obtained from remote sensing of the territory of a sensed state,[13] the Soviet-led socialist countries expressed the view that it was necessary to formulate criteria distinguishing between remote sensing data available for free dissemination and data which could not be disseminated without the express consent of the sensed state.[14] In a similar vein, France proposed that data above a certain threshold of resolution (which was yet to be determined) should not be disseminated without the consent of the sensed state.[15] Also, some years ago, the Soviet Union proposed that a resolution finer than 50 meters should not be disseminated without such consent.[16]

More recently, the Soviet Union expressed concern about the possibility that

> a number of receiving stations might in the future belong to private corporations or individuals. States should be responsible for everything connected with remote sensing activities otherwise the latter could become a source of confrontation and conflict. If receiving stations were in private hands and the operators of those stations felt free to treat the data they received as a marketable commodity [in the Soviet Union's view] a very different and unpleasant situation could arise. It was therefore necessary to establish some general principles defining the responsibilities of governments in that regard and work out a set of rules for the dissemination of remote sensing data.[17]

In contrast to the Soviet position the United States believed that any viable set of principles should foster the continued development of remote sensing programs and not inhibit their practical operation. In the view of the United States

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[11]Doc. A/AC.105/C.2/SR.383, p. 8 (1983).

[12] *Id.*

[13] *Id.*

[14]Doc. A/AC.105/C.2/SR.387, p. 2 (1983).

[15]Doc. A/AC.105/C.2/SR.384, p. 6 (1983).

[16]Doc. WG. III (1979)/WP.1.

[17]Doc. A/AC.105/C.2/SR.384, p. 8 (1983).

> the principles on remote sensing must respect the right of all states to conduct remote sensing programs in outer space in accordance with Article I of the Outer Space Treaty. Dissemination of data from civil remote-sensing programs could not be made dependent upon the prior consent of the sensed state since it would impair the usefulness and availability of such data. The principles could not assert the concept of state sovereignty over information obtained from outer space inasmuch as data, analyzed information, and technical information might be subject to proprietory rights that had to be respected. While the United States accepted international responsibility for the outer space activities of its governmental and nongovernmental entities, it could not accept the extension of such responsibility to terrestrial activities, except in accordance with generally recognized principles of international law.[18]

Another view stressed that

> a balance between the two types of factors was needed. The first type of factors were those effecting the freedom to disseminate data obtained by remote sensing and analyzed information based on those data. That freedom promoted the development of remote sensing programs and scientific progress in the field. The second group of factors were concerned primarily with the need to respect the national interests of the sensed states and not to harm their interests on the pretext of the freedom to disseminate data. But they also included both the undeniable right of the sensed state to have timely access to data before they were made available to third states and the responsibility of the state engaged in remote sensing, whether that was done by governmental organizations or by nongovernmental bodies.[19]

In addition to the foregoing views, the observation was made that

> certain states were in a dominant position which threatened to deepen the cleavage between "data rich" and "data poor" countries. Therefore, it was essential to bring about greater equality between technically advanced countries and the others. All countries should have an opportunity to participate in remote sensing activities through international cooperation, and remote sensing data should be as freely accessible as possible since a restrictive regime would favor the growing domination by the sensing states which already have data on all countries. At the same time adequate assistance to developing countries was needed to enable them to enjoy the benefits of remote sensing and interpret and apply data themselves.[20]

In a similar fashion the importance of a step-by-step approach was also advocated to fill the legal vacuum in the area of remote sensing, without prejudicing the development of legal instruments to keep pace with technical performance improvements incorporated into future satellites.[21]

B. International Law and the Use of Nuclear Power Sources (NPS) In Outer Space

Consideration of the use of NPS in outer space and the norms of international law applicable to it came to the fore following the crash on Canadian territory of the Soviet

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[18]Doc. A/AC.105/C.2/SR.387, pp. 8-9 (1983).

[19]Doc. A/AC.105/C.2/SR.384, p. 5 (1983).

[20]Doc. A/AC.105/C.2/SR.382, p. 4 (1983).

[21]Doc. A/AC.105/C.2/SR.383, p. 4 (1983).

Union's COSMOS 954 satellite which carried a nuclear reactor on board.[22] Canada and several other countries favored the idea of supplementing the existing norms of international law while the Soviet-led socialist countries were of the view that it was necessary to make a comprehensive study of relevant international law first.

There were a number of issues which surfaced during the early discussions. Many of them fell into two categories, those involving assistance and those pertaining to liability.

Issues in the first category related to the meaning of "necessary assistance," the launching state's right to participate in search and clean-up operations, the methods of determining the extent and duration of such operations, and the steps immediately to be taken by the affected states. Additionally, the access to the affected state's territory by search groups of assisting states, the extent of local experts' participation, the affected state's right to request assistance from a third state, and the ways of determining the methods of removing debris from the territory of the affected state were considered.[23]

With respect to the issue of liability, some countries expressed the view that liability for damage arising as a result of search and clean-up operations not conducted by the launching state cannot be imposed upon the launching state. As to the so-called "direct" and "indirect" damage, it was pointed out that these concepts were not incorporated in the Liability Convention of 1973. While some countries expressed the view that the special characteristics of nuclear power sources warranted the development of additional specific liability rules, other countries were of the view that the obligation of the launching state for consequences of its NPS use, together with the relevant provisions of the Outer Space Treaty and the Liability Convention, provided adequate bases for resolving virtually all of the issues.[24]

Following the recent entry into the earth's atmosphere of the component parts of the Soviet Union's COSMOS 1402 satellite which carried a nuclear reactor on board, concern was expressed by a number of countries that there was an urgent need for internationally accepted safety regulations providing for the same radiation limits adopted by the International Commission for Radiological Protection in connection with the terrestrial use of NPS. It was also stated that provisions should be made for assistance in case of an accident so that countries which were unable to protect themselves could get assistance on request. While several countries favored the idea of placing a moratorium on putting satellites with nuclear reactors in orbit around the earth until internationally agreed legal regulations on the use of NPS in outer space had been adopted, other countries opposed such a moratorium as neither feasible nor desirable, in particular, since this idea did not include the use of radioisotope sources which qualitatively had the same biological effect.[25]

In the course of the 1983 meeting of the Legal Subcommittee, a Canadian Working Paper consolidating some of Canada's earlier as well as more recent ideas expressed the desirability that each launching state provide information concerning its use of NPS and

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[22]For a detailed legal discussion of the issues raised by the Cosmos 954 accident, see 6 J. SPACE L. 107-170 (1978).

[23]Gorove, *How High is the Sky and Other Cosmic Questions*, 12 THE BRIEF 38 (Feb., 1983).

[24] *Id.*

[25]Doc. A/AC.105/318, p. 13 (1983).

that such use meet generally accepted international standards for radiological protection.[26] It also proposed a specific format for notification prior to anticipated re-entry of a space object containing NPS and requirements to offer all necessary assistance to states likely to be effected by such re-entry. The Working Paper stated *inter alia* that the launching state was liable to pay compensation for all damage caused by NPS, including all reasonable expenses for search and clean-up, and damages related to measures taken to prevent and limit radiation exposure and related to the number of people exposed and the degree of exposure.[27]

Following extensive discussions, the Legal Subcommittee's Working Group agreed that in the event a space object is malfunctioning with the risk of re-entry of radioactive materials to the earth, the launching state should provide timely information to the states concerned and to the Secretary General of the United Nations on system parameters and on the radiological risk of NPS.[28]

C. Issues Relating to the Definition and/or Delimitation of Outer Space and the Geostationary Orbit

While the question of definition and/or delimitation of outer space has been with us since the beginning of the space age, the additional issue pertaining to the geostationary orbit has become the subject of considerable discussion following the Bogota Declaration of 1976 in which eight equatorial countries claimed exclusive jurisdiction and sovereignty over segments of the geostationary orbit lying above their national territories.[29]

The basic issue with respect to the definition and/or delimitation of outer space has been whether to establish a precise demarcation line between air space and outer space and, if so, where such line should be located. The issue has been addressed both from a "spatial" and a "functional" viewpoint. Those advocating the "spatial" approach were in favor of establishing an easily determinable boundary between air space and outer space at a certain altitude above sea level. In this connection some delegations expressed support for the Soviet proposal that had set the boundary at an altitude not exceeding 110 kilometers above sea level and also provided for passage, at lower altitudes, through the air space of one state for space objects of another state for the purpose of reaching orbit or returning to earth so long as such passage caused no adverse effect in the territory of the state whose air space was crossed.[30]

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[26]Doc. A/AC.105/C.2/L.137, reproduced in Doc. A/AC.105/320, pp. 25-28 (1983), *infra,* pp. 172-175.

[27] *Id.* at 27. Another Working Paper by the Federal Republic of Germany made recommendations for the notification prior to re-entry of a nuclear-powered satellite and dealt with the experience from the re-entry of the COSMOS 1402 nuclear-powered satellite emphasizing the need for timely notification and comprehensive information to be given by the launching state to reduce the concern over the re-entry of a satellite with NPS on board. (Doc. A/AC.105/C.2/L.138 of 28 Mar. 1983, reproduced *infra,* pp. 175-179.)

[28]For details, *see* Doc. A/AC.105/320, p. 23 (1983).

[29]For a text of the Bogota Declaration of December 3, 1976, *see* 6 J. SPACE L. 193 (1978).

[30]Doc. A/AC.105/C.2/L.139.

Some countries which were in favor of the spatial approach stressed the importance and urgency of resolving the issue of definition and/or delimitation of outer space. They pointed out that different regimes apply to outer space and air space and therefore it was desirable to have a global and easily determinable boundary. They drew attention to the differences between the two regimes with respect to sovereignty which involved political and security considerations and sensitivities. They stated that the definition and/or delimitation based on altitude was essentially a legal and political, and not a purely scientific or technical, matter.[31] They were not in favor of the functional approach since in their view such an approach would lead to the applicability of two different regimes to the same geographical area and would also result in the weakening of the principle of national sovereignty over air space. A resolution of the problem based on altitude would prevent the occurrence of disputes and would facilitate international cooperation without impeding technological development.[32]

In the course of the discussions, the view was also expressed that a customary rule of international law had in fact developed, as states had now accepted the area above the lowest possible perigee of satellites as constituting outer space. A definition along these lines could be easily ascertainable and provisions could be made for the passage of space objects through air space.[33]

In contradistinction to the "spatial" method some countries favored what may be characterized as a "functional" approach, pointing out that it would be more productive to direct efforts toward the establishment of regulations to avoid possible interferences among space activities and adverse consequences for human life on earth. They stressed that such an approach would be for specific purposes rather than general as implied in the spatial approach and would better serve small states whose space objects, for geographical reasons, would more likely have to transverse the air space of another state.[34]

In line with the functional approach, it was also brought out that a definition and/or delimitation of outer space was not necessary nor feasible at the present time inasmuch as the development and application of space law had proceeded satisfactorily without it and additionally, because there was no scientific basis for such a definition and/or delimitation. The latter could only give rise to difficulties which might hamper space technological developments.[35] Additionally, it was stated that the existing outer space treaties were in fact based on a functional approach and a spacial definition would establish a vast, clearly defined area of air space over which states would not generally have the means to enforce their sovereignty.[36] It was also pointed out that air space and outer space were not distinguished by boundaries but by different activities and,

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[31]Doc. A/AC.105/32), p. 8 (1983).

[32] *Id.*

[33] *Id.*

[34] *Id.* at 9.

[35] *Id.* at 8.

[36]Gorove, *supra* note 23, at 9.

therefore, the future study of the definition of outer space should include the definition of outer space activities.[37]

A third view, which has been characterized as "pragmatic" was also advanced to the effect that a boundary between air space and outer space would not be responsive to any practical need now evident and could have unforeseen negative effects on the progressive development of space activities and space law.[38]

With respect to the issue of the geostationary orbit, a number of developing countries expressed the desirability of formulating regulations governing use of the orbit which they regarded to be of a *sui generis* character and a limited natural resource whose use would soon become saturated. They pointed out that the equatorial states had a special physical relationship with the geostationary orbit which necessitated the establishment of a special juridical regime. They recalled the decisions of UNISPACE 82 Conference as well as the 1982 Plenipotentiary Conference of the International Telecommunication Union in Nairobi. They expressed the view that the question relating to the geostationary orbit was not considered at the time of the drafting of the Outer Space Treaty and that not all states were parties to the treaty. Furthermore in their view continuation of the "first come, first served" practice would place the less developed states at a disadvantage. It was the technological advancements that underscored the pressing need for the promulgation of new legal regulation and appropriate technical planning which would take into account the needs of the developing countries and the special interests of equatorial countries in the rational and equitable use of the orbit.[39]

Other countries, including the space powers, did not question the right of all countries to equitable access to the orbit on an efficient and economical basis but pointed out that it was difficult to define the orbit's limit because technological developments were continually expanding it. They pointed out that the special characteristics of the orbit were due to its relation to the earth as a whole and not merely to a relationship to the equator. In their view, the orbit issue was essentially a question of the utilization of the radio frequency spectrum and as the matter was under consideration by the International Telecommunication Union (ITU), it was inappropriate for the Legal Subcommittee to prepare regulations with respect to it. It was stressed also that continued technological advancements would increase orbital capacity and keep pace with the demand for services. In view of this, orbital positions should be granted in accordance with demand at a particular time but without barring access to those who apply later. Thus appropriate management of the orbit through the ITU, rather than long-term inflexible planning, was required.[40]

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[37] *Id.*

[38] Doc. A/AC.105/32, p. 9 (1983).

[39] *Id.*

[40] *Id.* at 10.

D. Assessment and Suggestion for a Possible Compromise

The preceding brief overview of the current space law issues before the United Nations reveals considerable differences of opinion in practically all of the areas under discussion. To the extent that such differences are rooted in fundamental political or ideological beliefs, such as freedom of information versus state sovereignty, they are likely to remain very difficult to resolve. This appears to hold true in the field of remote sensing where the major stumbling block is the question whether a state that carries out remote sensing activities of the earth could without the approval of the sensed state freely disseminate the acquired data or information to third states, international organizations, and public or private entities. Those championing free dissemination of information regard it as a fundamental human right whereas those opposing it predicate their case on the principle of sovereignty and their inalienable right to dispose of their natural resources and of information concerning such resources.

While the matter of formulating draft principles governing remote sensing of the earth from space has been on the agenda of UNCOPUOS for many years, it may be recalled that initially many states entertained the view that remote sensing should not be carried out without the consent of the sensed state. This position, although also based upon sovereignty, was subsequently not pressed due largely to the fact that it appeared contrary to the principle of freedom of exploration and use of outer space, a fundamental principle of the Outer Space Treaty of 1967. As a result there is no requirement of consent by the sensed state to remote sensing in the draft principles under consideration. It is important to emphasize this because it clearly reveals that if all states had at present the ability to engage in remote sensing activities then they would be able to get on their own the very same information that they get under a system of free dissemination which many states are objecting to. This observation, should it be correct,appears to point in a direction which could conceivably carry a potential chance for a compromise. Such solution could be predicated on the recognition that there is a possibility of someone taking unfair advantage of remote sensing data by satellite. Presumably, the invocation of the principle of sovereignty and the insistence by a number of states on the requirement of consent by the sensed state to the dissemination of data is to protect such state's vital,mostly economic, interests in their natural resources from possible harm. Such harm might result if the intentional,unfair use of remote sensing data resulted in the user gaining economic benefits to the ascertainable detriment (damage) of the sensed state.

The taking of unfair advantage of remote sensing data by a user in the indicated manner could be declared unlawful and states could undertake to pass appropriate legislation to punish offenders. The laws pertaining to unfair competition and unfair trade practices might conceivably provide some guidelines in dealing with such problems. The possibility or likely incidence of unlawful use of remote sensing data in the indicated manner or whether it has ever occurred is not known since no specific instances or accusations have come to the limelight. This does not necessarily mean that it could never occur and, in view of this, a specific ban on such activity may allay some of the general concerns and fears.

Insofar as NPS is concerned differences of opinion do not seem as far apart as in the field of remote sensing. Thus it can be somewhat more realistically hoped that eventual consensus will emerge regarding notification, assistance and other requirements involving the use of NPS by satellites in low earth orbit.

The resolution of the issue of definition and/or delimitation of outer space appears to be the least pressing at the present time. The development of what may be regarded as a customary rule of international law that regards the area where satellites orbit the earth as outer space appears to make a precise physical determination no longer of immediate urgency. Such determination may entail or revive the troublesome question of where airspace ends and also raise the issue of the precise status of the area between airspace and outer space, should airspace not extend to the lowest boundary line of outer space.

Finally the issue of equitable access to the geostationary orbit/spectrum resource may well have a better chance for resolution by consensus in the scientific/technical setting of the 1985 World Administrative Radio Conference which is expected to consider the problem in detail.